

Artificial Intelligence Discipline PPGCC - UNIFESP

Professor Didier A. Vega-Oliveros

Project Proposal Statement

This subject aims to provide students with a technical and scientific challenge to delve deeper into the area of Artificial Intelligence (AI). It is expected that the projects developed throughout the discipline will result in the initiative of relevant scientific publications.

Project Objective:

Carry out an AI research and development project that demonstrates methodological rigor, originality and theoretical or practical applicability, at the level *in the strict sense* postgraduate. The work must be structured with a view to preparing a scientific publication.

Types of Work Accepted:

1. **Tutorial:** Short course or presentation and didactic explanation of advanced concepts in AI. It can be introductory (for beginning undergraduate students) or in-depth (for specialists). It must present key concepts, terminology and practical examples to facilitate understanding of the topic (15 pages).
2. **Original Article (Research Article):** Original research with well-defined contributions, new methods or experimental analyzes in the area of AI. Must include research problem, methodology, results and scientific implications (8 pages).
3. **Review:** Critical synthesis of academic literature on the topic of AI, with detailed analysis of concepts, approaches and future directions. May include experimental tests for validation (15 pages).

Format and Deliveries:

- **Delivery 1 Proposal (Short Paper): 4 pages**
 - article format to be defined.
 - Introduction: Clear presentation of the topic and motivation.
 - Gaps identified: Problems not yet resolved in the area.
 - Related works: References to previous research.

- Proposed materials and methods: Dataset and planned methodological strategy.
- State of the art: Existing technologies and limitations.
- Expected contribution: Proposal and how the work contributes to the area.

- **Delivery 2 (Final Article):** presentation and complementary material – article format to be defined.

- Proposal review: Recap of the problem addressed.
- Detailed methodology: Detailed description of the approach used.
- Experimental results and evaluation: Analysis of the experiments, evaluations carried out, or presentation of the topic with practical examples from the tutorials.
- Conclusions and scientific impact: Discussion of the contributions and impact of the work.
- Presentation of the project in the discipline. Presentation times to be defined.
- **Reproducibility:**
 - **Tutorial:** Codes and practical exercises
 - **Review:** Experimental tests on the state of the art and attachments.
 - **Original Article:** Repository with code along with results

Final Submission:

All final project work must be submitted to a pre-archive repository (*Zenodo*, *TechRxiv*) and should inform the DOI.

Assessment Criteria:

Short Paper (Weight: 30%)

- Clarity and definition of the proposal
- Theoretical foundation
- Quality of academic writing
- Methodological coherence
- Scientific relevance

Final Article (Weight: 50%)

- Originality of research or critical synthesis
- Quality of experiments/results
- Compliance with scientific standards
- Methodological rigor
- Code reproducibility and documentation

Presentation Assessment Criteria (Weight: 20%)

1. Clarity and Structure of the Presentation (20% of the grade) The organization of ideas and the fluidity of the presentation are assessed, including:
 - a. Logical and coherent organization of topics
 - b. Clear introduction of the topic, problem and objectives
 - c. Smooth transitions between presentation sections
 - d. Conclusion well defined and connected to the proposal
2. Quality of Slides (20% of the grade): The design, clarity and structure of the visual materials presented are evaluated:
 - a. Clean, readable design with good use of images and graphics
 - b. Slides with little text, prioritizing keywords and visual elements
 - c. Slide structure aligned with the presentation script
 - d. Adequate insertion of references and credits for materials used
3. Mastery of Content and Explanation (35% of the grade): The depth of knowledge demonstrated and the capacity for scientific communication are assessed:
 - a. Clear and in-depth explanation of concepts and methodology
 - b. Ability to answer audience and teacher questions confidently
 - c. Connection to relevant literature and project contribution to the field
 - d. Originality of the approach and clarity about its scientific relevance
4. Communication and Engagement Skills (15% of the grade): The posture and ability to interact with the public are assessed:
 - a. Clarity in speech, adequate rhythm and correct use of technical language
 - b. Eye contact and interaction with the public
 - c. Use of practical examples or analogies to facilitate understanding
 - d. Professional, confident and confident posture during the presentation
5. Presentation Time (10% of the grade): Respect for the stipulated time and its distribution throughout the presentation are assessed:
 - a. Length of presentation per student to be DEFINED.
 - b. Question session: up to 5 minutes after the presentation
 - c. Balanced distribution of topics, avoiding excessive concentration in a single section

Question Session: up to 5 minutes after the presentation

Slide Format: Digital presentations (PowerPoint, Google Slides, LaTeX Beamer or similar) with presentation PDF delivery.

Suggested Publication Vehicles:

Newspapers:

- Journal of the Brazilian Computer Society
<https://journals-sol.sbc.org.br/index.php/jbcs>

- Brazilian Journal of Applied Computing
<https://seer.upf.br/index.php/rbca/about/submissions>
- Journal of Theoretical and Applied Informatics
<https://seer.ufrgs.br/index.php/rita/about>
- The Latin American Center for Computer Studies CLEI
<https://clei.org/en/clei-electronic-journal-en/>
- Brazilian Journal of Informatics in Education
<https://journals-sol.sbc.org.br/index.php/rbie/about/submissions>
- More options from the Brazilian Computing Society (SBC)
<https://journals-sol.sbc.org.br/index.php/indice>
<https://sol.sbc.org.br/index.php/anais/workshops>
- Others

Conferences and Workshops:

- SBC events and workshops <https://sol.sbc.org.br/index.php/anais/conf>
- IJCAI, NeurIPS, COLT, UAI, others.

Final Observation: The use of LLMs as the main source of content will not be accepted. The work must demonstrate real effort, well-founded research and own development, ensuring effective learning and relevant scientific contributions. Rigorous application of methods, own development and in-depth scientific reflection are expected.

Suggested Themes for the Project:

Tutorial:

1. Introduction to LLMs and their applications in AI
2. Fine-tuning in Deep Learning models
3. AI, ethics and algorithmic justice
4. Intelligent agents and MCP/n8n protocol
5. Hyperparameter and architecture optimization
6. Ensembles with LLMs
7. Prompt engineering, RAG e variantes
8. Evaluation of open-source AI tools
9. AI applied to Education and intelligent tutors
10. Accessibility and inclusive AI

Research Article:

1. Supervised and self-supervised hybrid models
2. Anomaly detection or prediction in time series
3. Medical diagnosis with deep learning
4. Disease Prediction with GCNs and Diffusion
5. Efficient neural networks for embedded devices

6. Efficient neural networks in IoT
7. AI models for personalized medicine
8. Detection of extreme meteorological events
9. Interpretable and explainable models for sensitive data
10. AI applications for detecting and propagating epidemics, analyzing social networks, classifying nodes or graphs, predicting links, among others.

Datasets:

- paperswithcode.com/datasets
- kaggle.com/datasets
- Other datasets

Review:

1. State of the art in XAI
2. Fairness in AI: algorithms and applications
3. Comparison between LLMs and tuning techniques (RAG, few-shot, prompt engineer)
4. AI applied to ethics and algorithmic justice
5. Transfer learning in deep learning
6. Knowledge Bases for LLMs
7. Transformer Architectures for NLP
8. Generative AI and qualitative assessment
9. Security and adversarial attacks in AI
10. Responsibility and regulation in AI